

supporting policy reforms, new regulation, social innovation, behavioural change, new technology adoption and the integration of the gender dimension.

Scope: R&I results and new scientific knowledge can inform policy making and help address societal challenges through new technological and societal solutions, as well as through providing policy recommendations and policy options with a proper (scientific) understanding of the underlying conditions. Testing policy recommendations and findings of scientific research through experiments and novel methodologies increases understanding of the implications, risks and opportunities of possible new solutions, enabling societies to respond faster and more effectively to crises and build resilience and social cohesion.

This action aims at increasing the visibility and fostering the use of R&I results with high policy relevance to contribute to the recovery and resilience of Europe, while engaging research communities and citizens in an “experimentation space” for new, science based, socially inclusive and gender responsive policy initiatives and solutions.

The action will match policy relevant findings for EU resilience stemming from research (such as from EU H2020, HE and previous programmes as well as other EU programmes, and national level publicly funded research) with national, regional and local needs, and carry out the experimentation phase in co-creation with the research communities and citizens. At the core of the action is the design of appropriate trial mechanisms that will test research findings which can be translated to policy and new societal solutions, and thereby provide policy makers and citizens with a high degree of confidence and trust in responding to new challenges.

The project consortium may consider links with related knowledge management activities by the European Commission, including the European Commission’s Knowledge Centres and the Knowledge for Policy Platform hosted by the Joint Research Centre.

DEEPENING THE EUROPEAN RESEARCH AREA

OPEN SCIENCE

Proposals are invited against the following topic(s):

HORIZON-WIDERA-2022-ERA-01-40: Stepping-up institutional and territorial changes towards open and responsible research and innovation

Specific conditions	
<i>Expected EU contribution per project</i>	The Commission estimates that an EU contribution of around EUR 5.00 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.
<i>Indicative budget</i>	The total indicative budget for the topic is EUR 5.00 million.
<i>Type of Action</i>	Coordination and Support Actions

Horizon Europe - Work Programme 2021-2022
Widening participation and strengthening the European Research Area

<i>Eligibility conditions</i>	The conditions are described in General Annex B. The following exceptions apply: Legal entities established in non-associated third countries may exceptionally participate in this Coordination and support action.
<i>Legal and financial set-up of the Grant Agreements</i>	The rules are described in General Annex G. The following exceptions apply: Beneficiaries may provide financial support to third parties. The support to third parties can only be provided in the form of grants. The maximum amount to be granted to each third party is EUR 60.000. The respective options of the Model Grant Agreement will be applied. Beneficiaries should refer to General Annex B of the Work Programme for further information and guidance. Beneficiaries will be subject to the following additional dissemination obligations: Beneficiaries must make proactive efforts to freely share, in a timely manner and as appropriate, all relevant results with the other grants awarded from this work programme part subject to the same additional dissemination obligations. Beneficiaries must acknowledge and incorporate these obligations in the proposal, outlining the efforts they will make towards meeting them, and in Annex I to the Grant Agreement.

Expected Outcome: Projects are expected to contribute to the following expected outcomes:

- Consolidated evidence base on sustainable institutional and territorial changes towards open and responsible research and innovation;
- A central point of expertise providing services and support for research and innovation institutions, and territories, to open up to society;
- A significant number of institutions and territories in the European Research Area (ERA) become more porous to society, align with the needs, values and expectations of society, improve the excellence of their research and innovation, promote gender equality, and reduce instances of ethical misconduct;
- Reduction in disparities between institutions and territories in terms of their attention to open and responsible research and innovation.

These targeted outcomes in turn contribute to medium and long-term impacts:

- Open and responsible research and innovation practices mainstreamed throughout territorial and institutional settings;

- Increased impact of EU R&I outputs and the conversion of knowledge and ideas into products and services.

Scope: Research and innovation institutions play a key role in creating an enabling environment for opening research and innovation towards society, sharing research outputs, improving research integrity and gender equality, and promoting science education, societal engagement (such as citizen science and other forms of co-design and co-creation) and two-way science communication. At territorial level, interactions between different institutions from across the quadruple helix ⁷⁹, and including societal actors such as civil society organisations and non-governmental organisations, are essential to ensuring that the processes and outcomes of research and innovation are aligned with the needs, values and expectations of society.

However, European research and innovation actors lack a consolidated evidence base and guidance based on EU investments in this area to date to support them, there are significant differences in attention to different aspects of responsibility and openness between institutions and territories, rewards and incentives at the workforce and institutional level are often misaligned or even disfavour openness to society, and there is no central point of reference or provider of services and expertise to turn to.

This action consists of three parts, all of which must be addressed:

The first involves consolidating the evidence base and develop innovative guidance and other materials, which can support institutions and territories to implement sustainable institutional changes and open up to society⁸⁰. It should take into account and build on the learning and knowledge developed by Horizon 2020's Science with and for Society programme ⁸¹ and potentially other sources of knowledge.

The second part involves financial support to third parties, by launching 'cascading grant' call(s) to support institutions and territories from across the ERA to implement sustainable institutional changes towards open and responsible research and innovation. This may require consultancy or other kinds of support services to be developed and rolled out to successful institutions. A significant number of institutional changes (e.g. 70-100 individual institutional changes) should be expected in beneficiary organisations and territories. As such, a significant proportion of the funding should be allocated to the 'cascading grant' mechanism. One or more call(s) for proposals should be launched, which could focus on specific disciplines,

⁷⁹ A model of cooperation between industry, academia, civil society and public authorities, with a strong emphasis on citizens and their needs.

⁸⁰ Institutional changes are characterised as: 1) a change to how the institution governs/structures itself, 2) expected to have meaningful impact within the institution concerned, 3) intended to last beyond the lifetime of funding (i.e. not one-off activities). See <https://op.europa.eu/s/olV2>.

⁸¹ Particularly relevant projects to build upon were supported by: ISSI-5-2014, GARRI-1-2014, ISSI-5-2015, SwafS-04-2016, SwafS-05-2017, SwafS-05-2018-2019, SwafS-14-2018-2019-2020, SwafS-23-2020. Moreover, this action should build on the evidence base for benefits developed by the MoRRI and Super_MoRRI projects. Projects focused on implementing Gender Equality Plans under SwafS (including those supported by topics GERI-4-2014, SwafS-03-2016-2017, SwafS-09-2018-2019-2020) may also be of relevance. The RRI-Tools project, from FP7's Science in Society programme, may also be relevant: <https://rri-tools.eu/>.

MoRRI country clusters⁸², or other pertinent criteria, but with the underlying goal of reducing disparities between institutions and territories in terms of their attention to different dimensions of openness and responsibility.

The third part involves acting as a central point of expertise and support services on open and responsible research and innovation for institutions and projects under Horizon Europe and within the European Research Area. This point of expertise should have appropriate high-level visibility, and the ability to interact and support all parts of the research and innovation system (all parts of quadruple helix, disciplines, sectors). In order to achieve the expected outcomes, international networking is advised to support co-operation on the issue at global level and provide expertise in support of institutions and projects.

The action should evaluate its impacts and develop recommendations useful to policy makers and those responsible for the governance of research and innovation institutions. The action should raise awareness of the benefits of open and responsible research and innovation to organisations across the ERA. It should develop close co-operation with other relevant projects, with a view to fostering collaboration and the early sharing of knowledge and evidence.

HORIZON-WIDERA-2022-ERA-01-41: Increasing the reproducibility of scientific results

Specific conditions	
<i>Expected EU contribution per project</i>	The Commission estimates that an EU contribution of around EUR 2.00 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.
<i>Indicative budget</i>	The total indicative budget for the topic is EUR 4.00 million.
<i>Type of Action</i>	Research and Innovation Actions
<i>Legal and financial set-up of the Grant Agreements</i>	The rules are described in General Annex G. The following exceptions apply: Beneficiaries will be subject to the following additional dissemination obligations: Beneficiaries must make proactive efforts to freely share, in a timely manner and as appropriate, all relevant results with the other grants awarded from the same call topic. Beneficiaries must acknowledge and incorporate these obligations in the proposal, outlining the efforts they will make towards meeting them, and in Annex I to the Grant Agreement.

Expected Outcome: Projects are expected to contribute to the following expected outcomes:

⁸²

<https://op.europa.eu/s/olV4>

- Structured understanding of the underlying drivers, of concrete and effective interventions - funding, community-based, technical and policy - to increase reproducibility of the results of R&I; and of their benefits;
- Effective solutions, policy-, technical- and practice-based, to increase the reproducibility of R&I results in funding programmes, in communities and in the dissemination of scientific results;
- Greater collaboration, alignment of practices and joint action by stakeholders to increase reproducibility, including but not limited to training, specialised careers and guidelines for best practice.

These outcomes should contribute to medium and long-term impacts:

- Increased proportion of reproducible results from publicly funded R&I;
- Increased re-use of scientific results by research and innovation;
- Greater quality of the scientific production.

Scope: Reproducibility refers to the possibility for the scientific community to obtain the same results as the originators of a specific finding. As such, reproducibility is core to scientific progress, and there is debate on whether there is a ‘crisis of reproducibility’ in contemporary science. At a time when funding levels for R&I are under scrutiny globally, and societal trust in the outcomes of research and innovation become increasingly essential, there is a need to address inefficiencies in the research process, to avoid useless and costly repetition, to maximise return on investment in R&D&I, to prevent the propagation of mistakes, and to facilitate the translation of results into innovations.

Therefore, this topic aims to fund activities to

- a) determine how increased reproducibility generates gains and savings in the R&I process and improve overall performance - alongside the demonstrated positive effects on their quality, integrity and trust-worthiness, and
- b) find, experiment and mainstream concrete solutions and best-practices to increase the reproducibility of research funded with European taxpayers’ money, including through the more systematic integration of sex and gender as variables whenever relevant.

Consequently, actions should help understand and promote reproducibility by:

- 1) creating an open knowledge base of results, methodologies and interventions on the drivers and consequences of reproducibility for the R&I system; and to fill the main gaps in such knowledge;
- 2) develop, validate, pilot and deploy practices and practical tools for funders, publishers and scientists;